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**AI-POWERED SYSTEM FOR PHISHING
ATTACK DETECTION**

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AI-Powered System for Phishing Attack Detection

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SUPERVISOR APPROVAL

AI-POWERED SYSTEM FOR PHISHING ATTACK DETECTION

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The thesis was prepared under the supervision of the project supervisor, Dr. Noormadinah binti Allias. It was submitted to the Faculty of Computer and Mathematical Sciences and was accepted in partial fulfilment of the requirements for the degree of Bachelor of Computer Science (Hons.) Netcentric Computing.

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JANUARY 5, 2025

STUDENT DECLARATION

I certify that this thesis and the project to which it refers is the product of my own and that any idea or quotation from the work of other people, published or otherwise are fully acknowledged in accordance with the standard referring practices of the discipline.



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ABSTRACT

The prevalence of internet-based communication in the present digital age has predisposed users to phishing attacks, which have now taken a more advanced form of SMS (smishing) and malicious URLs in addition to the straightforward email scam. The conventional security systems like blacklist-based and rule-based are becoming irrelevant in dealing with such new changes in attacks because the systems fail to respond to new phishing attacks. Despite being more adaptive, machine learning methods suffer due to the high-dimensionality of data samples with irrelevant and redundant attributes that tend to increase computational costs and decrease classification performance. This paper will concentrate on the selection of features to detect phishing in three mediums of communication, namely email, URL, and SMS. The publicly available datasets of labelled phishing and legitimate samples were used, which was followed by the use of data preprocessing tools such as tokenization, normalization, and Term Frequency Inverse Document Frequency (TF-IDF) to transform the textual data into numbers. The most significant feature subsets applicable in the detection of phishing were discovered using a nature-inspired metaheuristic algorithm known as Binary Harris Hawks Optimization (BHHO). The features were then tested on machine learning classifiers to estimate their performance when it comes to enhancing detection performance. Accuracy, precision, recall, F1-score, and confusion matrix analysis as a scientific method of experimental evaluation indicate that BHHO is effective in the initial step of dimensionality reduction of features and the final step of boosting the accuracy of classification. There is a positive indication in the results that BHHO can be utilized as an optimized feature selection method in phishing detection and play a role in more efficient and effective machine learning-based cybersecurity solutions.

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LIST OF ABBREVIATIONS

ANN	Artificial Neural Network
BHHO	Binary Harris Hawks Optimization
CNN	Convolutional Neural Network
DT	Decision Tree
GA	Genetic Algorithm
GB	Gradient Boosting
KNN	k-Nearest Neighbors
LR	Logistic Regression
NB	Naive Bayes
RF	Random Forest
SVM	Support Vector Machine
TF-IDF	Term Frequency-Inverse Document Frequency

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The Internet has become a necessity to many people in this digital age as they use it to do everything. Individuals tend to provide personal information, including passwords, bank account and personal data, and other private information that only he/she is aware of (Ahmad et al., 2024). The attackers have also devised improved ways of attacking individuals, largely due to the fact that majority of the individuals have become overly reliant on the digital tools. The most popular and dangerous form of cyberattack is phishing.

Phishing is no longer just another type of email fraud but a significant cybersecurity threat that involves spamming via SMS (smishing) and websites (Saranya et al., 2025). Attackers exploit the weaknesses of the user, deceiving the user to disclose sensitive data through fake URLs, fraud messages and malicious websites (Siva et al., 2024). Posing as someone the user knows and trusts and deceiving the user to disclose his sensitive information is one of the tricks, which are commonly employed by the attackers.

In the past, phishing attacks, e-mails, were easier to detect since the mistakes and the design were too apparent. Then, with the help of conventional means, a blacklist is employed to prevent the attack. Nevertheless, since the phishing techniques are more advanced, it has outsmarted the traditional, rule-based security protocols, and more advanced solutions are needed. Therefore, the old approaches to cybersecurity are not capable of adjusting to the emerging phishing schemes because of the nature

of their limitations. Hence, it is evident that the sector requires intelligent and dynamic detection systems.

The majority of cybersecurity practitioners have switched to Artificial Intelligence (AI) and machine learning. In the case of traditional machine learning, much manual effort is needed to identify important data characteristics (Alshingiti et al., 2023). In the meantime, big data where the dimensions of data are large and the relevant features are numerous can slow down the system and decrease the accuracy of the classifier.

The weakness of the existing technology presents a big gap, which is swift technologies, power technologies capable of handling various types of information simultaneously and deliver precise outcomes. In response to these problems, there is a rising trend where AI (Artificial Intelligence) and machine learning methods are applied to cybersecurity research. Even though machine learning-based methods are more adaptable, they tend to deal with high-dimensional data with irrelevant and redundant features, which may decrease the accuracy of the detection task and complicate the process of computation (Alshingiti et al., 2023). Consequently, effective feature selection has been a very essential feature of phishing detection systems.

1.2 Problem Statement

Current Machine Learning-based phishing detection models frequently process a large number of features from datasets without proper optimization (Alshingiti et al., 2023). As a result, the problem can cause irrelevant and redundant features, thus introducing noise into the data, which leads to increased computational costs, potential overfitting and influences the model accuracy (Danasingh et al., 2020).

The process of feature selection is essential in preprocessing machine learning because it enhances the performance of the model, decreases the cost of computations, and simplifies the model itself by prioritizing the input features that are the most useful

(Mohammed and El-Feky, 2025). Moreover, there are three broad categories of feature selection techniques namely, filter-based methods, wrapper-based methods and embedded methods. Firstly, filter-based approach is a standalone algorithm that identifies the significant features based on the statistical ranking including information gain, Chi-square test, Fisher score and correlation coefficient without relying on any learning protocols (Kaur et al., 2021). Second, as Kaur et al. (2021) explain, the wrapper-based approach is used to select the most suitable set of features by trial and error with the learning algorithms through search methods like sequential selection and heuristic search. Finally, embedded method achieves feature selection process during the training and integrates both the selection and learning process to enhance efficiency and accuracy.

The filter-based methods are effective yet these methods tend to retain a collection of characteristics which might not be the optimal to classify, whereas the wrapper-based methods are computationally inexpensive since they rely on the repeated training of learning models (Mohammed and El-Feky, 2025). In the case of embedded methods, the chosen characteristics are skewed to a single model, despite lower chances of overfitting (Syeda and Asghar, 2023). These shortcomings reveal that the current feature selection tools are possibly not adequate to process high dimensional data and complex data. Hence, there is a need to identify a feature selection model that has a high model accuracy and low computational expenses.

1.3 Research Objectives

The main objective of this study is to investigate the ability of the proposed feature selection for phishing detection. To achieve this goal, the following objectives are outlined:

1. To identify the important features for phishing detection using Binary Harris Hawks
2. To compare the performance of Binary Harris Hawks with the traditional feature selections.

3. To evaluate the effectiveness of the selected features in improving phishing detection accuracy using machine learning classifiers.

1.4 Research Scope

This research scope is aimed at creating phishing detection system, which can be used to analyze three communication channels, these are emails, URL and SMS. The research is based on feature selection process through Binary Harris Hawks Optimization (BHHO) so as to determine the most useful features in the process of phishing detection. The data is preprocessed and classified to aid in the assessment of the features chosen. Besides, the study uses publicly available datasets with labeled phishing and legitimate samples.

1.5 Research Significance

The paper advances research literature by discussing how the optimized feature selection can be used to improve phishing detection. Savyanavar et al. (2024) recognize that the conventional feature selection approaches have numerous limitations, which comprise irrelevant or redundant information, so they provide significant recommendations to researchers in future phishing detection and data processing research. The results may be consulted by other researchers that want to be more efficient in choosing features applied to cybersecurity systems.

Moreover, the phishing detection systems are also enhanced in terms of their efficiency and accuracy, as this research is aimed at the most significant aspects, as well as minimizing superfluous data and costs of calculations (Danasingh et al., 2020). The suggested methodology is applicable in the real-world detection systems with the use of emails, URL and SMS that can assist the organization to come up with more robust and effective cybersecurity solutions.

CHAPTER 2

2.0 LITERATURE REVIEW

2.1 Overview of Phishing

Phishing is an internet-based crime in which criminals impersonate trusted organizations in order to access confidential information such as usernames, passwords or financial data. It may occur through alternative mediums like email, URL or text messages (Ahmad et al.). To gain the confidence of the users and scam them into giving out their information, attackers usually develop phishing resources and sites that resemble an authentic site (Joseph and Srinivasan, 2025). Thus, the intricate structure of counterfeit links and interfaces complicate the process of identifying threats in the case of users unless they possess sufficient awareness or defense system (Ahmad et al., 2024).

The primary purpose of a phishing attack is to exploit the human vulnerabilities and the psychological tendencies to fool the users into divulging personal data (Korkmaz, 2026). Moreover, over the years phishing tactics are becoming more sophisticated and complex such that they are challenging to identify using traditional security measures (Ahmad et al., 2024). Traditional detection systems like blacklists and rule-based systems rely on the static sources of attack signatures and fixed rules (Doshi et al., 2025). Consequently, these practices struggle to adapt to the ways of the modern and zero-day threats which speaks volumes of the urgency of detecting systems that are adaptive, intelligent and automated (Saranya et al., 2025).

2.2 Types of Phishing Attacks

2.2.1 Email-Based Phishing Attacks

Email phishing refers to the act of sending fraudulent messages via email that come from reliable sources such as banks, service providers and coworkers, in which to trick the users into clicking a malicious link or giving their sensitive information like credit card numbers or passwords (Ramprasath et al., 2023). Emails are commonly used with persuasive language, urgent messages and inserted links or files to manipulate users (Chanis & Arampatzis, 2024). Phishing emails can be detected through features such as suspicious sender details, mismatched domains, abnormal subject headings and language patterns in the message (Zhang et al., 2025). Choosing relevant email features is a must to enhance the accuracy of the phishing detection model.

2.2.2 SMS-Based Phishing Attacks

SMS based phishing, also known as smishing, is an attack that target mobile phone of the user through text messages in which the text message can pretend to be a legit message. The text messages commonly contain shortened links or urgent messages to triggers user in act quickly (Mishra & Soni, 2020). The previous research shows that short text messages in short forms or symbol forms and text messages consists of idioms and abbreviations are characteristics of the smishing attacks (Mishra and Soni, 2021). Smishing attacks are difficult to detect because of the feature limitations and the lack of sufficient real datasets.

2.2.3 URL-Based Phishing Attacks

Attackers use malicious links to redirect users to fraudulent sites that are designed exactly like legitimate platforms (Ewasakul & Oransirikul, 2024). The links

can be spread through emails, SMS or online advertisements. Moreover, detecting URL phishing involves analyzing features such as the length of the URL, special characters, suspicious term and abnormal domain patterns (Kustiawan & Ghauth, 2025). As phishing URLs change rapidly, choosing the most important features is vital for enhancing detection accuracy and reducing complexity.

2.3 Types of Feature Selection

Feature selection techniques are commonly categorized into three types, which are filter-based methods, wrapper-based methods and embedded methods. Each type of feature selection methods has different characteristics, advantages and limitations that will be discussed below.

2.3.1 Filter-based methods

Based on Mandal et al. (2021), filter-based methods are a technique that selects the most important features from the datasets without relying on other classification models. The key features are chosen based on the statistical significance to support the classification or prediction of the data. One of the advantages of filter-based methods is in text classification and sentiment analysis, where dimensionality is reduced by ranking and choosing the most significant terms to enhance both accuracy and clarity (Mohammed & El-Feky, 2025). The limitation of this model is assuming that the important features do not change between datasets, which reduces adaptability in real-world applications (Mohammed & El-Feky, 2025).

2.3.2 Wrapper-based methods

According to Yin et al. (2023), wrapper-based methods are a technique that evaluates feature subsets based on actual performance in a given machine learning model. Wrapper methods are not like filter methods that are independent, it can rely on different classifications or prediction models. Each of the selected feature subsets requires the model to train and test, which results in higher computational costs and

longer processing time. The advantage of using a wrapper-based method is that the method considers how features depend on each other while interacting with the classifier and it is not highly demanding in terms of computation (Kaur et al., 2021). As for limitation is, methods carry the risk of overfitting but deliver good results and require more time to complete. (Kaur et al., 2021).

2.3.3 Embedded methods

Embedded methods select useful features during training, which combines selection and learning in a single process to enhance efficiency and accuracy (Kaur et al., 2021). The advantages of embedded methods are that the methods can perform variable selection, feature selection and classification together during the training process (Kaur et al., 2021). This method depends on a classifier, which can result in different feature selection based on the classifier's assumptions (Syeda & Asghar, 2023).

2.4 Machine Learning Approaches

Random Forest (RF) is frequently used in detecting phishing, as multiple predictions are more accurate and reduce overfitting due to its use of multiple decision trees. It is able to efficiently process URL based features without need of access to web content as well as when combined with deep learning methods like CNN to extract the features, it will process images with accuracy rates above 99 percent (Yang et al., 2021). Kavya and Sumathi (2024) state that structured feature information, i.e. URL metadata and domain information, is also processed by RF, allowing an all-encompassing analysis of phishing indicators in the process. Random Forest has been viewed as the appropriate model to be used with intelligent phishing detection systems because it has better generalization and is quicker to execute as compared to the performance of classifiers, such as ANN, SVM, C4.5, and Rotation Forest (Keerthana et al., 2025).

Moreover, XGBoost or Extreme Gradient Boosting is an ensemble algorithm, which integrates several weak learners to produce a stronger model, therefore, improving predictive performance (Gu and Xu, 2022). According to Saraswathi et al. (2024), XGBoost is able to work with large datasets and intricate patterns successfully, which is why it can be considered a credible tool when it comes to detecting phishing. It is also frequently quite successful in classification tasks.

In addition, K-Nearest Neighbors (KNN) is a simple algorithm that uses training data to classify unseen data based on a chosen distance measure (Gu & Xu, 2022). According to Kalikiri et al. (2025) KNN is a non-parametric method that categorizes data points based on how they are to one another, using majority vote among the nearest neighbors. It also often used to fill in missing values in datasets. The main idea is that points located near each other in the feature space are likely to be similar. Therefore, the algorithm predicts the class or value of a new point by looking at the attributes of its closest neighbors.

Moreover, Support Vector Machines (SVM) creates boundaries in high-dimensional space to separate different classes while keeping the widest possible margin between them. It works effectively when the classes are clearly different in their features making it useful for separating phishing URL from legitimate ones (Saraswathi et al., 2024). According to Ajibade et al. (2025), SVM are efficient in memory use because only the support vectors affect the decisions. They work well in high-dimensional spaces, even when there are more features than samples. However, they face difficulty with noisy or overlapping data and can become slow when handling large datasets. The accuracy also depends on choosing the right kernels and parameters so poor choices can reduce performance.

2.5 Related Works

2.5.1 Existing Methods of Feature Selections

The prior studies have been using a range of feature selection algorithm to improve the detection of phishing in email, SMS messages, URLs and websites. Mat Rani et al. (2023) used TreeSHAP and Information Gain and obtained the result of 98.59% accuracy with XGBoost; nevertheless, huge datasets may include the redundant or irrelevant features that are challenging to remove without the help of special tools. On the same note, Bahaghighat et al. (2023) applied PCA with constant feature elimination, in which XGBoost achieved 99.2 percent accuracy but conventional techniques have low processing speed and cannot handle constant features in practical data.

Dahr and Gaafar (2025) used hybrid feature selection options which embody a combination of header, body and URL feature in addition to stacking ensembles and obtained a high accuracy of 99.53% in email phishing detection, however high number of features can add more detection time as well as larger sample size. Siva et al. (2024) applied Canopy feature selection using hybrid classifiers to URL phishing, and the performance of their approach was high, but this heavily relies on the quality of preprocessing. Genetic Algorithms were used by Kocyigit et al. (2024) to solve URL-based features optimizing features to enhance efficiency and minimize overfitting; however, MA is prone to early convergence to a suboptimal solution. In a different work, Savyanavar et al. (2024) used ensemble feature reduction in phishing webpages with an accuracy of 96.57% whereas redundant or irrelevant features are hard to remove when using large datasets.

Overall, the existing feature selection methods enhance the accuracy of phishing detection, but they have issues such as high computational cost, dependence on a specific dataset, the possibility to lose valuable features, and preprocessing sensitivity. These drawbacks suggest the necessity of more adaptive and more robust feature selection approaches on phishing detection in the research.

Table 2.1 *Comparisons between feature selection methods*

Author(s), Year	Types of Phishing	Feature Selection Method	Classifier used	Main Findings	Limitations
Mat Rani et al. (2023)	Email, spear phishing, whaling, voice, and website phishing	Information Gain (IG) and TreeSHAP	Naïve Bayes (NB), Random Forest (RF), and XGBoost	XGBoost achieved the highest accuracy of 98.59%	Traditional non-URL features (web content/external) are time-consuming and require high computing power.
Bahaghighat et al. (2023)	E-banking, E-commerce, social media	SMOTEENN & dropping constant features	LR, KNN, NB, RF, SVM, XGBoost	XGBoost reached 99.2% accuracy	Traditional feature selection methods have low processing speed and struggle with constant features in real-world datasets.
Dahr & Gaafar (2025)	SMS Spam (Smishing) and Network Intrusion (DDoS)	Filter (Chi-Square), Wrapper (Lasso, RFE),	XGBoost	Filter (Chi-Square) was the most accurate (75.76%)	Traditional feature selection suffers from high execution costs, limited memory for large datasets, and may not select truly representative features.
Connolly & Atlam (2025)	Email Phishing	Hybrid feature selection	Stacking Ensemble: RF and XGBoost	Achieved 99.53% accuracy and 99.85%	High feature counts can increase detection time

			(base learners) with Logistic Regression (meta learner)	recall	and require larger sample sizes.
Siva et al. (2024)	Fake emails and fraudulent websites	Canopy feature selection with grid search and cross-fold validation	DT, LR, RF, NB, GB, KNN, SVM, and a Hybrid LSD model (LR+SVM+DT)	The Hybrid LSD and Stacking Classifiers achieved 100% accuracy	No feature selection method is foolproof; effectiveness depends on quality of preprocessing.
Savyanavar et al., 2024	Email, Smishing, Vishing, Whaling	Ensemble reduction (IG, Chi-Square, Gain Ratio, Correlation)	Random Forest	Achieved 96.57% accuracy	Large datasets contain redundant/irrelevant features that are hard to eliminate without specialized methods.
Kocyigit et al., 2024	URL-based phishing	Genetic Algorithm (GA) with Local Optimization	Random Forest, XGBoost, and ANN	GA reduced features to fewer than 45	GA may converge prematurely to suboptimal solutions if population diversity isn't maintained.

2.6 Summary

This chapter review gives the key ideas and the current research concerning the phishing detection systems. It begins by introducing phishing, describing why phishing and how phishing evolves with time. Various channels of phishing were explained like email, SMS and URLs to exhibit the range of methods employed by attackers and how they are difficult to detect. Next, addressed the approaches to feature

selection, including filter-based, wrapper-based and embedded. All of the methodology were described with respective strengths and limitations particularly when dealing with large and high dimensional data. In the section, it was pointed out that to maximize accuracy of detection and reduce redundancy of data, appropriate features ought to be chosen. Some of the machine learning algorithms that are being used widely in the detection of phishing were also examined in the chapter, namely, Random Forest, XGBoost, K Nearest Neighbors, and Support Vector Machines. The talk was devoted to the way these algorithms work and their efficiency in separating phishing and legitimate data. Finally, the past studies on feature selection to phishing detection were reviewed and compared. Although most of the existing approaches have yielded good results, they still face some challenges that included reliance on specific datasets, loss of key features, and limited flexibility. These weaknesses imply the necessity of elaborating a more efficient strategy of feature selection that is discussed in the proposed study.

CHAPTER 3

3.0 RESEARCH METHODOLOGY

3.1 Introduction

This study applies a quantitative research approach because it focuses on the data that can be measured and expressed through numbers. It evaluates performance using metrics such as accuracy, precision, recall and F1-score. The study can provide a transparent evaluation of the hybrid model and traditional methods. Additionally, the experimental research is also applied where the models are trained, tested and evaluated systematically in a monitored environment using datasets.

3.2 The Methodology Framework

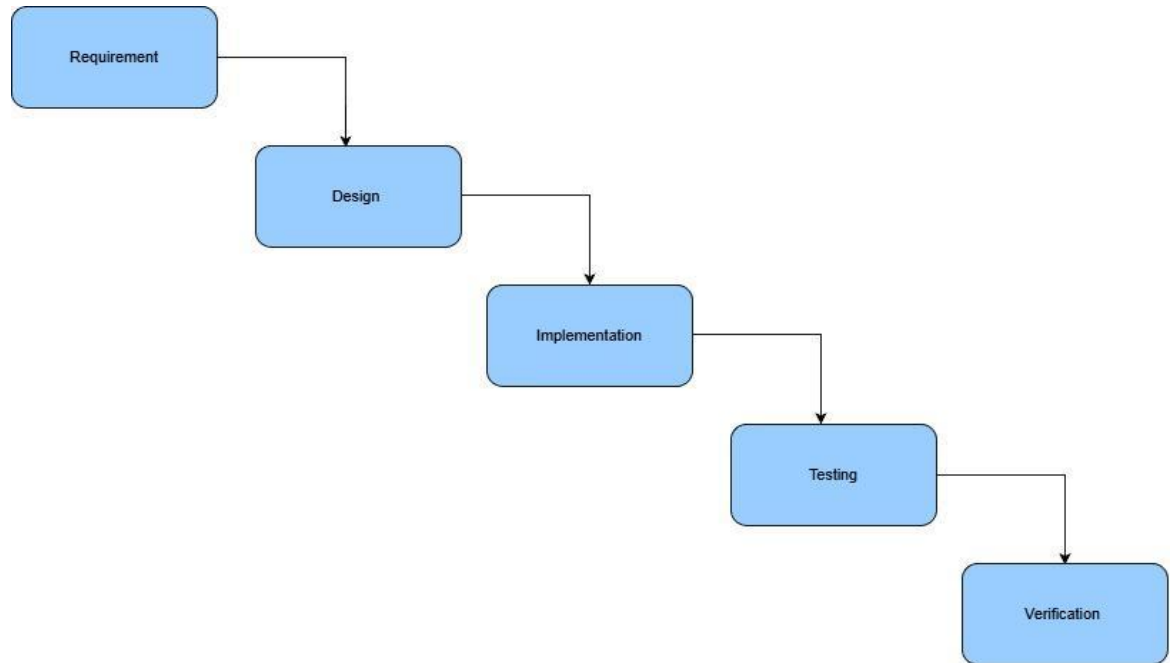


Figure 3.1 The Waterfall Model

In the requirement phase, objectives, scope and limitations of the research is identify and defined. Many researchers have set a clear objective for detecting phishing in emails, URL and SMS and emphasize non-functional requirements such as improved accuracy and efficiency (Siva et al., 2024). Based on Mughaid et al. (2022), URL-based systems evaluate large datasets using performance metrics accuracy, precision, recall and F1-score. Meanwhile, email-based system needs to classify whether the email is phishing or legitimate. Email-based system uses the same metrics but include the false-positive rate.

The overall system structure is planned in the design phase. This involves designing a data preprocessing workflow. The choice of feature selection methods and classification techniques is specified to ensure the system fulfills the requirements. This phase defines the workflow for cleaning and transforming raw data, identifying

relevant features, enhancing model performance and selecting algorithms (Kadulkar et al., 2022).

This implementation stage is involved in the implementation of the system as per the design requirements. The preprocessing data is implemented using the selected software tools that include feature selection attributes and model development of the Random Forest, Support Vector Machine, K-Nearest Neighbors, Artificial Neural Network, Deep Neural Network, Convolutional Neural Network, XGBoost and model combinations on phishing data (Mughaid et al., 2022).

The implemented system is performance evaluated by testing on measures of metrics accuracy, precision, recall and F1-score in a variety of algorithms and feature sets (Gualberto et al., 2020). Several configurations are experimented to ensure reliability of systems and to meet the goals of the research.

The tasks done in this stage of verifications include; confirming that the development system is functioning according to the requirements and research objectives set. The results are evaluated and justified through comparison with the already existing methods. This stage makes sure that the solution suggested is efficient, stable, and applies to the detection of phishing (Gualberto et al., 2020).

3.3 System Diagram

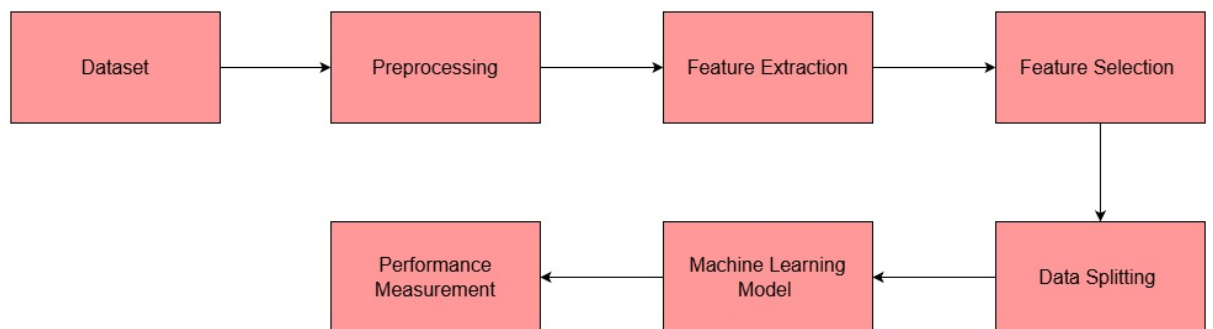


Figure 3.2 The System Workflow

The following workflow of this phishing detection system begins with collection of datasets that are of three sources namely, email, SMS and URL datasets. Phishing information and valid data are included in these datasets to serve as an input to the system. UCI Machine Learning Repository, URL-Phish and publicly available text message are gathered as phishing and legitimate samples respectively. PhiUSIIL phishing URL data set at the UCI Machine learning Repository has 235,795 data points (134,850 legitimate and 100,945 phishing URLs). Besides, 54 features were extracted which are representing lexical and structural properties (UCI Machine Learning Repository). The URL-Phish data includes 111,660 distinct URLs having 22 extracted lexical and structural attributes and was gathered in PhishTank and alternative trusted sources (Mendeley Data). The structural characteristics of this dataset are the length of URL, the length of domain, subdomains number, ratio of digits, entropy, and the use of HTTPS. SMS phishing database is collected based on publicly available text messages. This data set contains 5971 text messages which were classified as Legitimate or Smishing. It contains 489 spam, 638 smishing messages and 4844 ham messages.

The stage of data preprocessing is that in which raw data must be cleaned and normalized before proceeding to the further analysis. The initial step is turned to small letters to exclude the problem of case sensitivity. Noises are then eliminated by deleting punctuations marks, numbers, special characters and unnecessary symbols which are irrelevant data. This is followed by tokenization which is used to separate the text in the URL or email into useful segments like domain names and path segments or words whose feature extraction algorithms (TF-IDF) can be used to extract the significant textual patterns. Also, the stemming or lemmatization process transforms words into root words and removal of stop-words to remove frequent but not informative words when searching email contents. Through these preprocessing, there is a better consistency of the data; and also, the dimensionality is reduced and the quality of features to be used in the subsequent stages is bettered.

Term Frequency-Inverse Document Frequency (TF-IDF) method is applied to carry out feature extraction after data preprocessing. TF-IDF transform the textual data that is cleaned into numerical feature vectors by indicating weights to words according

to their significance in a document, as well as in the entire dataset. The TF IDF representation gives emphasis to the phishing keywords which are characteristic of phishing email such as verify, urgent and account suspended which are frequently used in phishing emails but are less frequently used in legitimate emails. In the case of phishing sites, character-based TF IDF features obtained with noisy HTML code and plain text have been useful in finding subtle patterns, without necessarily using external services and thus the zero-hour phishing sites are identified with the same high accuracy. In the same breath, the application of TF IDF to the URLs of logins is able to distinguish lexical variations that distinguish phishing pages and valid logins pages, which when applied in the real world can offer a visual identity that is similar, but a login page with ill intentions. Accordingly, all documents are encoded as a high dimensional place that gives the relative importance of the textual elements contained in them.

The feature selection method used is Binary Harris Hawks Optimization (BHHO). BHHO is inspired by the cooperative hunting behavior of Harris hawks, and is formulated to find a set of optimal features in a binary features space. In this the hawks are associated with the potential feature sets that can be obtained based on the TF IDF feature where one is used to indicate the feature is selected and a zero is used to indicate the feature is not selected. In the process of optimization, a random initial collection of hawks is created, and each of them is evaluated by a fitness function that considers both the accuracy of classification and the count of features selected. The algorithm switches between the exploration and exploitation phases to be able to search the feature space. During the exploration phase, BHHO explores different combinations of features to avoid early convergence and on the exploitation stage, it focuses on narrowing down on promising combinations of features to improve the quality of the solution. BHHO gradually converges to an optimal set of features which increases the detection accuracy and also minimizes dimensionality through repeated updates. Once the ideal feature set has been received, the trimmed-down feature set is used in subsequent workflow phases. After the optimized features the data splitting process is used to separate them into training and testing data. These data sets are utilized in the training and testing of the machine learning classifier. The inclusion of BHHO into the working process implies that a classification with the use of only the

most significant and informative characteristics is used, which increases accuracy, minimizes overfitting, and improves computational efficiency.

It is followed by the division of the dataset into both training and testing sets after the feature selection operation by BHHO. The conventional ratio is 80:20 and 60:40 according to Mohamad Asraf Daniel et al. (2025), which are applied to refer to the training and evaluation stages. The training data consists of diverse collection of the labelled examples of legitimate and phishing. The datasets are meant to document the different attributes and trends that are attributed to phishing attacks. Thereafter, the testing datasets are designed in such a way as to test the capacity of the model in dealing with unfamiliar data. This procedure makes sure that the model is reliable and it is less prone to overfitting and it also provides a more accurate estimation of phishing detection.

The process of machine learning model is introduced after splitting the data. Random Forest algorithm is not a feature extraction algorithm, but a classification model. All textual data are processed and transformed to numerical form with the use of TF-IDF method prior to reaching the classifier. The Random Forest then trains itself on these feature vectors together with their class labels in order to know how to part phishing and real cases during training. It constructs many decision trees using random subsets of the data, and each tree constructs its own rules based on the various combinations of TF-IDF features in order to be able to distinguish phishing data and valid samples. Every decision tree generates individual classification upon making a prediction. The majority vote in the trees is used to determine the final result, and the combine approach enhances the reliability, as well as the accuracy of phishing detection.

Finally, the measurement of the phishing detection system performance is conducted through normal classification metrics. A confusion matrix is created to describe the outcome of prediction in the context of a true positive, true negative, false negative and false positive. According to these values, such evaluation metrics as accuracy, precision, recall and F1 score are obtained. Accuracy is the total correctness of the predictions, precision the ratio of the phishing instances that are hit on, recall

the capacity of the system to recognize the real phishing cases and F1 score is a fairly balanced measure of precision and recall. The use of various measures of evaluation can be used to get a holistic view of the system performance especially in cases of imbalanced data sets.

3.4 Performance Evaluation Metrics

Four conventional evaluation metrics will be used to determine the effectiveness of the proposed performance outcomes and the Binary Harris Hawks Optimization (BHHO). The metrics are quantitative levels of the effectiveness of the model to make a distinction between phishing attacks and keep the false alarms to a minimum. As stated in the article by Alshingiti et al. (2023), the accuracy, precision, recall and confusion matrix were the metrics used to assess the prediction results in order to evaluate the overall performance.

- Accuracy: The accuracy will be the percentage of correctness in prediction, indicating the extent to which the system recognizes the cases of phishing and legitimate ones (Alshingiti et al., 2023).

$$Accuracy = \frac{TP + TN}{TP + TN + FN + FP}$$

- Precision: The number of phishing webpages, which are automatically classified as phishing, is referred to as precision (Alshingiti et al., 2023).

$$Precision = \frac{TP}{TP + FP}$$

- Recall can be assessed as the degree to which the model finds and identifies all authentic threat of phishing in the dataset (Alshingiti et al., 2023).

$$Recall = \frac{TP}{TP + FN}$$

- This is a process of averaging of the highest accuracy and the highest recall rate to create a single measurement (Alshingiti et al., 2023).

$$F1 - Score = \frac{2 \times (Precision \times Recall)}{(Precision + Recall)}$$

3.5 Experimental Setup

3.5.1 Hardware Requirements

The computer which is used to execute the experiments has an AMD Ryzen 7 5825U processor. This CPU has a 16-thread architecture, 8 cores with a maximum boost clock of 4.5 GHz. The system has 12GB of DDR4 memory and has AMD Radeon of graphics as a source of integrated visual and parallel processing assignments.

Table 3.1 Hardware Requirements

No.	Hardware Components	Specification
1	System Model	HP Laptop 15-fc0xxx
2	Processor (CPU)	AMD Ryzen 7 5825U
3	Cores / Threads	8 Cores / 16 Logical Processors
4	Base Clock Speed	2.0 GHz (Max Boost up to 4.5 GHz)
5	Memory (RAM)	12 GB DDR4 3200MHz
6	Storage	512 GB NVMe SSD
7	Graphics (GPU)	AMD Radeon™ Graphics (Integrated)

3.5.2 Software Requirements

This part is on software requirements that were involved in the creation of phishing detection system. The following are the enumerated tools and platforms that are popular in this project.

Table 3.2 Software Requirements

Category	Software / Tool	Purpose
Operating System	Windows 10 / Windows 11	A platform used to develop and run the phishing detection system
Programming Language	Python 3.	Used for data preprocessing, feature selection, and classification
Development Environment	Anaconda Distribution	Manages Python environments and required libraries
Interactive Environment	JupyterLab	Used for data analysis, experimentation, and result visualization
Data Processing Libraries	NumPy, Pandas	Used for numerical operations and dataset handling
Machine Learning Library	Scikit-learn	Used for model training, testing, and evaluation
Visualization Libraries	Matplotlib, Seaborn	Used to visualize results and performance metrics
Dataset Format	CSV	Used for storing and processing phishing datasets

3.6 Gantt Chart

The Gantt chart was developed to indicate the project development over 28-week or approximately one year during an academic semester. This is to secure that each of the phases, beginning with the introduction up to the last documentation are coordinated and uniform in the development of the project.

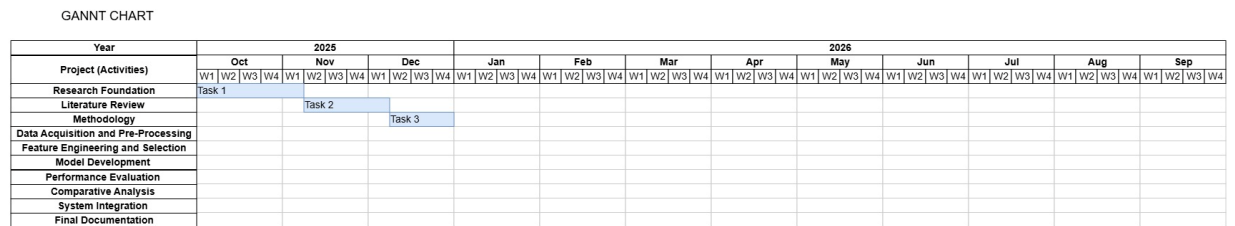


Figure 3.3 Gantt Chart

3.7 Summary

This chapter gives the research approach that was employed to construct the proposed phishing detection system. The research adhered to the Waterfall model in order to have a systematic and organized development process that included requirement analysis, system design, implementation, evaluation, and verification. The approach also described the general workflow, namely, dataset collection, data preprocessing, feature extraction with TF IDF and feature selection with the help of Binary Harris Hawk Optimization (BHHO). The feature selection method that utilizes BHHO is the key contribution of the present study and is used to handle the high dimensionality of TF IDF features and enhance the effectiveness of the classification process. The enhanced set of features was further divided into training and testing data, and the classification was performed using the Random Forest algorithm. The standard measures used in evaluating system performance included accuracy, precision, recall

and F1 score. This methodological construct provides a solid foundation of the experimental analysis and the outcome of the offered system will be justified and discussed in the following chapter.

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